

**UNIVERSITY OF ILLINOIS-CHICAGO
DEPARTMENT OF CHEMICAL ENGINEERING**

**CHE 397 – SENIOR DESIGN II
Spring 2020**

Lecture:

Section 1: T/Th: 9.45am – 11am

Section 2: T/Th: 1pm-2:15pm

No Classes:

March 24th and 26th (Spring break)

Office Hours:

Tuesdays and Thursdays 11am-1pm (Dr. Bilgin)

Instructor:

Dr. Betul Bilgin, bbilgin@uic.edu

Teach Asst:

Seyyedfaridoddin Fattahpour

Text:

Towler, G., Sinnott, R., Chemical Engineering Design: Principles, Practice and Economics of Plant and Process Design, Second Edition, Elsevier, 2012. (Textbook website: booksite.elsevier.com/Towler)

Course Description:

This course builds on the skills taught in CHE 396. Design methodology and principles will be applied to the design of large-scale chemical processes and plants. Design needs to be analyzed from process simulation, safety, environmental impact, controls, optimization, feasibility and economics. Competing processes must be critically analyzed and presented. Essential part of this course is working in teams effectively. Selection of design topic is structured to be flexible to open up opportunities of creativity. Since the course is structured to encourage flexible approaches to design problems, intermediate presentations and reports are required to keep students on track and to evaluate the feasibility of their design.

Course Objectives:

This course involves the application of all prior CHE concepts to the chemical process design. Design needs to be analyzed from safety, environmental impact, feasibility, risk and economics perspectives. Course aims to equip students with the skill to evaluate open-ended design problems conceptually, analytically and practically.

Learning Objectives :

At the end of the course, students should be able to:

- Develop design plan and project charter for each design phase
- Work effectively with team members
- Present effectively technical material based on audience
- Identify key components in market analysis and construct design basis
- Analyze competing processes
- Write technical content in a way that is appropriate to the audience
- Apply economic analysis
- Identify potential safety concerns in the process and formulate strategies to mitigate those
- Simulate full process with Aspen Plus or SuperPro
- Size process equipment

Requirements/Constraints:

- All the calculations need to be submitted in excel
- All the work need to be archived in blackboard and all the mentors will have access to them
- There will be 4 reports through the semester.
- There will be 4 presentations through the semester. During presentations, all the group members are expected to participate.
- Final report needs to be submitted to the mentors 2 weeks prior to the end of semester.
- 7 stepping stones for the project
 1. Process description
 2. Material/energy balances and PFDs
 3. More detailed equipment calculations (at this point, student are encouraged to contact with vendors)
 4. Pricing equipment, economics, cash flows
 5. Control scheme
 6. Safety/potential hazardous materials list
 7. Final report

Report/Presentation Deliverables:

Report 1:

- Introduction (your goal is to get the reader excited to read your report)
 - Clear definition of big problem statement
 - Why this problem is important?
 - Explanation of your goals and objectives
 - Market analysis
 - Clear explanation of judging criteria (more environmental friendly, more economical etc.)
- Process description (your goal is make the reader believe that you already did the research on possible ways to make the product, you analyzed them well and have chosen your way critically)
 - What is the process?
 - What are possible solutions? (competing processes (strengths/weaknesses of them), chemistries) (graphs, figures, tables would be helpful for comparison)

- What is your solution?
- Why you have chosen that solution?
- Appendix 1(design basis)
- Appendix 2 (BFD and material balance)
- Preliminary appendix 3 (PFD) (For report 1, it is only preliminary PFD)
- Preliminary appendix 4 (material balances) (For report 1, it is only rough material balance meaning that mass balance block in and out. Estimates of products, by-products, waste etc.)

Report 2:

- Final PFD
- Detailed mass balance (This means detailed inputs and outputs to each part of the process along with conditions ie, temperature, pressure, etc.)
- Equipment list (Most of the major equipment should be identified along with a good start on the equipment list)
- Equipment sizing (column sizes, pump and compressor sizing, heat exchanger sizing, along with horsepower requirements and heating/cooling loads) (One distillation tower sizing or separation unit sizing should be done by hand calculation. Hand calculation and Aspen results should be compared and analyzed critically)
- Preliminary utilities demand summary for entire plant.

Report 3:

- Process controls
 - Describe control objectives
 - What are the main process parameters?
 - What is your control strategy?
- Environmental and safety concerns
 - List environmental issues, air emissions, water discharges, etc.
 - What design features you would use to mitigate the effects of these discharges?
 - Outline the process hazards which you must legally and ethically address. (That means, for instance, all OSHA PSM highly hazardous materials should be listed, including quantities on site. Process hazards associated with toxic, explosive, hot and cold process streams should be identified and a safeguard included in the design.)
- Cost analysis
 - detailed cost analysis (NPV, sensitivity analysis, payback period, capital cost, Aspen econ analysis, etc..) Analyze your results rather than just listing the numbers.

Report 4 (Final Report):

- Corrections/updates of report 1,2 and 3.
- Utilities
- Executive summary
- Recommendations
- Heat exchanger network(optional)

Assigning groups

Every group will include 4-5 members.

Students will choose one person who they want to work with and one who they do not want to work with.

Dr. Bilgin will assign the groups based on student's choices.

Each group will be assigned to one mentor.

Responsibilities

-Each group is responsible to meet with their mentor weekly (physical meeting or online meeting or phone conversation).

-Each group is responsible to choose a group leader and rotate the leader among the members of the group every three weeks. Leader's responsibility is to set up the meetings with mentor.

-Before the meeting; each group is required to submit a meeting agenda to corresponding mentor 2 days before the meeting. If the agenda is not submitted, then meeting will be canceled.

-After the meeting; each group is required to submit meeting notes to corresponding mentor next day after the meeting.

-Presentations will be 15 min. There are 8 groups in morning section and 6 groups in the afternoon sections. Morning group presentations will be on Tuesday/ Thursday 10.30am-12.30pm. Afternoon group presentations will be on Tuesday/ Thursday 1-2.30pm as announced on the course blackboard site.

-Intermediate reports will be reviewed by professor.

-Final reports will be reviewed by assigned mentor of each group.

Course Outline:

Date	Concept	Assignment
January 14 th 2020	Syllabus review	
January 16 th 2020	Group work	
January 21 st 2020		
January 23 th 2020		
January 28 th 2020		
January 30 th 2020	NO CLASS	
February 4 th 2020	First presentation	
February 6 th 2020	First presentation	
February 11 th 2020		
February 13 th 2020		
February 18 th 2020		
February 20 th 2020		

February 25 th 2020		
February 27 th 2020	NO CLASS	
March 3 th 2020	Second presentation	
March 5 th 2020	Second presentation	
March 10 th 2020		
March 12 th 2020		
March 17 th 2020		
March 19 th 2020		
March 24 th 2020	NO CLASS	
March 26 th 2020	NO CLASS	
March 31 st 2020	Third presentation	
April 2 nd 2020	Third presentation	
April 7 th 2020		
April 9 th 2020		
April 14 th 2020		
April 16 th 2020		
April 21 st 2020		
April 23 th 2020	NO CLASS	
April 24 th 2020	EXPO	
April 28 th 2020	Final presentation(tentative)	
April 30 th 2020	Final presentation(tentative)	

Exams:

There will be not be any exam.

Attendance:

Students are expected to attend all classes, and attendance is incorporated as 5% of the final grade. Students are allowed two excused absences or one unexcused absence. If absences exceed these limits, the student will receive a zero for attendance.

Grading Policy:

Course grades will be weighted according to the following grading formula.

Attendance:5%
Presentation 1:10%

Report 1:	10%
Presentation 2:	10%
Report 2:	10%
Presentation 3:	10%
Report 3:	10%
Final Presentation:	12.5%
Final Report:	12.5%
Meeting with Mentors:.....	10%